SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	2023.	Date:	09/10/2026.

Code of Conduct

I madala virupaksha (192372205) *(Name / Reg No)* certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Virupaksha

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature:	Signature:	Signature:
Date: 09/10/26	Date:	Date:

88/100

spectrum auction for = 45MHz
= 45000KHz

25KHz for every simplex

full duplex = 2x25KHz = 50KHz

$\frac{45000\,KHz}{50\,KHz} = 900\,KHz$

560 total voice channels.

shared across 7-cell

for each cell $\frac{560}{7} = 80$ channels

carriers spacing at 200KHz and rolls out
100 carriers.

each carriers 8 time slots.

Total = 200KHz x 100 x 8
= 800 voice channels

radiate 80w, handset transmits 4w.

power control = $10\log_{10}(\frac{80}{4})$.
$= 10\log_{10}(20) = 13.05$ db.

carrier to interface ratio(C/I) = $-60dBm - (-92dBm) = 32dB$.

6. Total channels = $\frac{140}{\frac{420000KHz}{300KHz}}$, per cell = $\frac{140}{7}$
= 140 channels = 20 channels

7. engineer have receiver twice
n=4, power to receive = 64w.
received power = $\frac{64}{24} = \frac{64}{16} = 4$.

8. invert-square law. (P/R^2)
double radius = 2R.
transmit power = 23 = 8.4

9. cluster size = 4, voice channels = 15
Total count = 4 x 15 = 60.

10. vehicle travelling at = 100 km/h.
carried = 900MHz carriers
doppler shift. $(\frac{100}{3\times10^8}) \times 900\times10^6$.
= 83.61

11. sms divided into 8 time slots.
duration for single shot = $\frac{5}{8} = 0.625$ ms

12. allocated bandwidth = 24MHz
carriers spacing at = 200KHz.
carriers frequency = $\frac{24000KHz}{200Hz}$ 120.
= 120 carriers

13. transmits = 8w, signal = 33dB.
$10.\log_{10}(8000mw) = 39$ dB.

14. free-space path = 2.5km = 2500cm.
accessing. = 1900 MHz
1m as the reference.
free-space path = $20\log_{10}(2.5) + 20\log_{10}(1900) + 32.44$
= 129.9 dB.

15. consists of = 6 cell
each carry up = 250 calls at peak.
Total capacity = 6x250
= 1500

16. Little law $L = \lambda w$.
L = 4000 calls/s x 3min (180se)
= 4000x180
= 720,000 active calls.

17. Bandwidth = 150 x 25 KHz
= 3,750 KHz
= 3.75 MHz.

18. guard band = 40KHz
30 adjacent frequency bands.
total bandwidth = 40x(30-1) =
= 40x29
= 1,160 KHz

19. radius = 1.5 km, diameter = 3km
distance in hrs = $\frac{90km}{3km} = 30$.

20. no. of centers = 8
base station = 40
each station serving
= 250 users
Total capacity = 8x4x250
= 32x250
= 80,000

21. cell area $R^2 = (400m)^2$
= 160000m = 0.1659 km
cells required = $\frac{4005q\,km}{0.1659\,km}$
= 2500 cells

22. FDMA channels.
= $\frac{60000KHz}{300KHz}$
= 200 channels

23. 1200 subscribers
each = 4 calls/h
avg = 2.5 minutes.
TRAFFIC = $1200(\frac{1}{4}\times\frac{2.5}{60})$ 200/15 6
= 200 erlangs.

24. bandwidth = 40MHz
consumes = 400KHz
max users = $\frac{48800KHz}{400KHz}$
= 100 users.

(25) simultaneous users = 600
network comprises = 1200

Total = 600 × 1200 = 720,000 users

(26) paging sounds = $\frac{200 \text{ cells}}{4 \text{ cells/s}}$

= 50 sounds.

(27) offered traffic = $\frac{20{,}000 \times 3}{60}$

= 2000 erlangs.

(28) network policy is
15% of its 600000 subscribers.

roaming = 0.15 × 600,000
= 90,000 subscribers

(29) percentage = $\frac{3}{4} \times \frac{25}{100}$
= 75%

(30) channels per cell = $\frac{48\text{MHz} \times 200}{4}$

= $\frac{48000 \times 200}{4} = \frac{240}{4} = 60$

(31) Gsm channels = $\frac{300 \text{ MHz}}{200 \text{ KHz}}$

= $\frac{1500}{3000}$
$\frac{}{200}$

= 150 channels.

(32) succesful handoffs = 0.99 × 2500
= 2475 handoffs.

(33) Recieved power = 10 log10 (12000mw)
= 40.79 dbm

40.79m − 45 = −4.21 dbm

(34) 7-cell cluster, 900 user
capacity = 7 × 900
= 6300 users

(35) device move at = 15 km/h
connected to = 1.8 GHz

doppler shift duration = $\left(\frac{15/3.6}{3 \times 10^8}\right) \times 1.8 \times 10^9$

= 250 Hz

(35) 4.5 ms, split into 8 equal slots
duration = $\frac{4.5}{8}$ = 0.6 ms.

(37) 250,000 simultaneous calls
calls at a rate = 70,000 p/m
blocked chance = 4.666.67
calls out 250,000
= 0% blocked.

(38) cluster capacity = 9 × 600
= 5,400 users

(39) 60,000 cells per second
0.4% dropper
calls dropped every second =
0.004 × 6000
= 240 calls/sec

(40) transmit at 43 dBm
signal measure = −87 dBm
signal underson = 43 dBm − (−87 dBm)
= 43 + 87 = 130 dB

(41) station covers = 2.5 km²
150 identical station
Total are = 2.5 × 150
= 375 sq km

(42) Channels per call
= $\frac{48{,}000 \text{ KHz}/200 \text{ KHz}}{6}$

= $\frac{240}{6}$ = 240 channels.

(43) pagging cycles
= $\frac{150{,}000 \text{ users}}{6000 \text{ users}}$
= 25 cycles

(44) Handoff dosing cell = $\frac{9 \text{ min}}{3 \text{ in}} - 1$
= 3 − 1 = 2 handoffs

(45) SNR = −85 dBm − (−115 dBm)
= −85 + 115 = 30 dB

(46) minimum bandwidth = 250 × 150
= 37,500 kbps.

(47) generate traffic = 1500 × $\left(\frac{4 \times 2.5}{60}\right)$
= 250 erlangs.

(48) control channel bandwidth
= 250 × 12 KHz = 3000 KHz

(49) Total network = 1500 × 600
= 900000 kbps (900 mps)

(50) area of 1 towers = $\pi \times 4^2$ = 50.27 sq
Towers needed = $\frac{350}{50.27} \approx 7$ towers.